

Name: _____



Defining features of linear relationships

Task A: Numerical features of linear relationships

1) These are tables of values for linear relationships. Fill in the missing values and complete the statements about the **rate of change**.

a)

x	-2	-1	0	1	2	3	4
y			2	7			

For every change of +1 in x there is a change of ___ in y

b)

x	-2	-1	0	1	2	3	4
y			-4		10		

For every change of +1 in x there is a change of ___ in y

c)

x	-2	-1	0	1	2	3	4
y			2				8

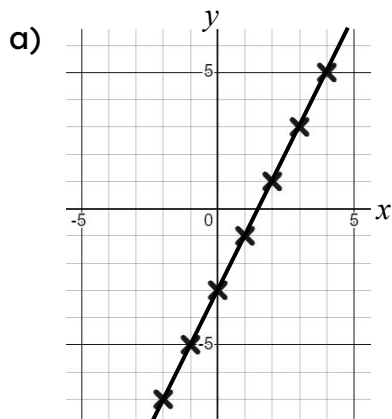
For every change of +1 in x there is a change of ___ in y

d)

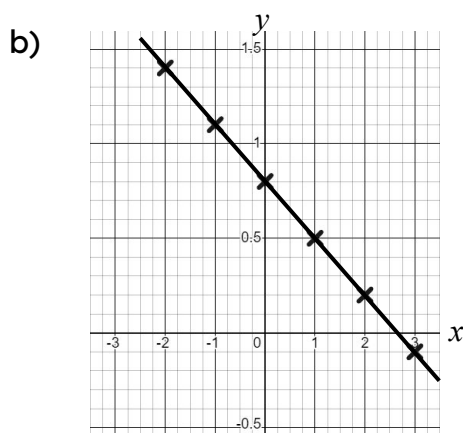
x	-2	-1	0	1	2	3	4
y			12			-3	

For every change of +1 in x there is a change of ___ in y

2) Use the graph to complete the table and write a sentence about the **rate of change**.



x	-2	-1	0	1	2	3	4
y							



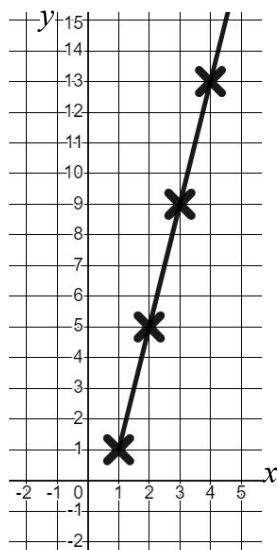
x	-2	-1	0	1	2	3
y						

Task B: Graphical features of linear relationships

1) Izzy has completed a table of values and plotted coordinates for the linear relationship $y = 4x - 3$

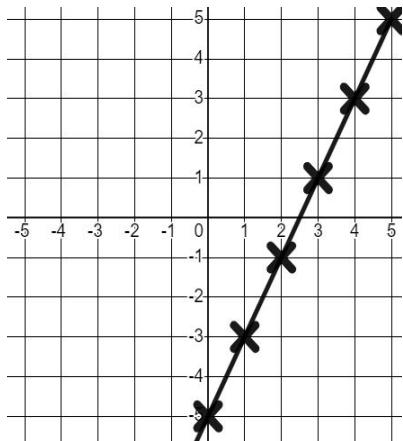
x	1	2	3	4
y	1	5	9	13

The line $y = 4x - 3$ starts at 1 and has a **rate of change** of +4 each time.

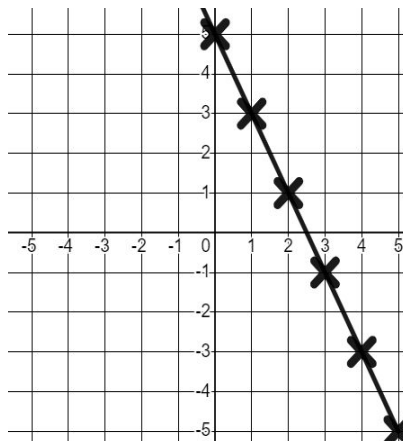


2) Match the graphs to their unique descriptions.

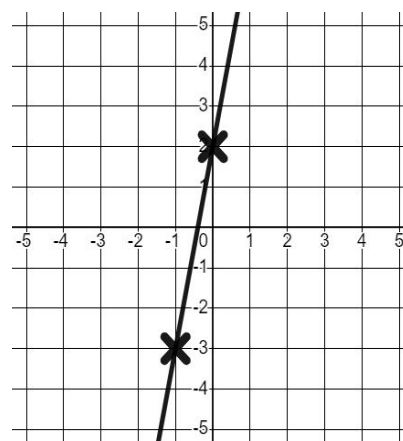
Intercepts the y -axis at $(0, 5)$, **rate of change** of -2



Intercepts the y -axis at $(0, 2)$, **rate of change** of $+5$



Intercepts the y -axis at $(0, -5)$, **rate of change** of $+2$



3) Write unique descriptions for these two graphs. Be sure to include the words y -intercept and rate of change..

